Lying to a Bayesian: Optimal Randomisation in Repeated Pricing

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Abstract

Reinforcement Learning has succeeded at a wide range of games (Chess, Go, Atari). Here we apply RL methods to a game-theoretic problem in economics : a repeated pricing game between a Bayesian-updating producer and a consumer with private willingness-to-pay. We established existence of the mixed strategy Nash equilibrium for the continuous-action extension through Reny (1999), which we then approximate computationally via three RL methods. We have used three different approaches (Monte Carlo, Q-learning, Deep Q-Network) to simulate the benefits of randomisation for consumer's behaviour. All three approaches show a consistent result that $\epsilon^* < 1$, supporting randomisation as an obfuscation strategy. Tabular Q-learning shows no clear monotonic trend in $\epsilon^*(\beta)$, while DQN recovers the monotone decrease (Spearman $\rho = -0.428,\ p = 10^{-4}$) predicted by Monte Carlo, reconciling the previously-observed discrepancy. We show this discrepancy is not due to state discretisation - enriching the tabular state to (μ, σ) does not recover the trend.

1 Game Setting

Here I define a infinitely repeated game between consumer and producer.

- **Consumer**

  Consumer has willingness to pay \$100.00 for a product, so his profit is the difference between his wtp and the price charged by the producer. Consumer's action space is $\{\pm 1\}$ where 1 represents take the producer's offer, while -1 represents reject the offer. Consumer's objective function is a sum of all expected future profits, i.e.

$$V_0(a) = \max_{a\in\{\pm1\}}((100 - x_0)\mathbb{1}(a = 1) + \beta E[V_1(a)])$$

the above is the consumer's bellman equation to be optimised where $\beta \in [0,1]$ represents the time discount factor.

Consumer's policy :

$$\pi_t^c(a_t|x_t) = \epsilon\mathbb{1}(a_t = +1, x_t \leq 100) + (1-\epsilon)1/2\mathbb{1}(a_t \in \{\pm1\})$$

where $\begin{cases} \pi_*(a_t|x_t) = \begin{cases} 1 & x_t \leq 100 \\ -1 & otherwise \end{cases} \\ u_t(a_t, x_t) = (100 - x_t)\mathbb{1}(a_t = +1) \end{cases}$. Here the baseline (deterministic) policy is

$$\bar{\pi}(a_t|x_t) = \mathbb{1}(a_t = +1)\mathbb{1}(x_t \leq 100) + \mathbb{1}(a_t = -1)\mathbb{1}(x_t > 100)$$

– Information hiding (Reputation Theory)

  : if consumer always accepts when $p \leq 100$, and refuses when $p > 100$, producer perfectly infers $w^* = 100$ and always charges $x = 100$ forever, giving consumer 0 surplus. Randomisation creates ambiguity.

– Exploration-Exploitation (Bandit Theory)

  : consumer trades off immediate surplus (exploiting good prices) against long-term surplus (maintaining uncertainty). The optimal ϵ^* trades these off, similar to the UCB's exploration bonus.

Then we have a consumer's value function

$$V(x_t) = E_{x_{t+1}\sim\mu, a_t\sim\pi_t(\cdot|x_t)}[u_t(a_t, x_t) + \beta V_{t+1}(x_{t+1})|\pi_t, x_t \sim \pi_t(\cdot|x_t)]$$

- **Producer**

Producer produces a product at the cost of \$50.00. The producer observes consumer's behaviour given each price and updates their belief about consumer's willingness to pay. Then they aim to extract full profit by setting the price as the consumer's willingness to pay.
Producer assumes consumer's willingness to pay is drawn from an unknown

distribution. The producer adopts Gaussian prior for WTP for tractability: $\mathcal{N}(\mu_t, \sigma_t^2)$ and charges the posterior mean $x_t = \mu_t$,which by a law of large numbers concentrates on the true WTP under the truthful play; it updates (μ_t, σ_t^2) via Bayesian updates after each observation.

- at $t \in \{0, \ldots, 6\}$

  the producer explores consumer's choice by playing "take a middle" policy.
- $t \geq 7$

  the producer uses data to fit the normal distribution and get the mean and set the price as the updated mean.

Here I denote by μ a measure for the producer's distribution for the states (the state here means the price the producer sets).

Producer's policy :

$$\pi_t^p(x_t) = \begin{cases} \pi' & 0 \leq t < \lfloor \log_2(H-L) \rfloor \\ \mu_t & t \geq \lfloor \log_2(H-L) \rfloor \end{cases}$$

where μ_t is the mean of normal distribution updated from the past t-observations, and π' is the optimal policy for a "guess a number" game;

Algorithm 1 optimal policy π'

$t = 0$
  while True **do**
    For a interval $[L_t, U_t]$, $x_t = \lfloor (L_t + U_t)/2 \rfloor$
    if $a_t = +1$ **then**
      $L_{t+1} = x_t$
    else
      $U_{t+1} = x_t$
    end if
    if $L_{t+1} \neq U_{t+1}$ **then**
      $t+ = 1$
    else
      break
    end if
  end while

Here I assume the range [0,150], hence $\lfloor \log_2(H-L) \rfloor = 7$. Thus, for time $t \in \{0,\dots,6\}$, producer explores consumer's behaviour by playing "take the middle" policy. At $t = 7$, the producer fits data into the normal distribution with mean μ_7 and standard deviation σ_7, i.e. $\mu \sim N(\mu_7, \sigma_7^2)$. After $t = 7$, the producer updates his/her distribution by Bayesian updates.

2 Theoretical Foundations

2.1 Baseline Model

My baseline is that consumer plays truly to their preferences; if the price is higher than their willingness to pay, they reject it and conversely the price is lower than or equal to the willingness to pay, they accept it. Then it comes "guess-a-number" game to the producer, and they get maximum profits once they get the right number, while consumer only gets 0 surplus forever. In that game, the optimal policy is taking the middle number. In this setting, the producer sets price in range [0,150], the game will be

$$\begin{array}{rcl} (x_0, a_0) & = & (75, +1) \\ (x_1, a_1) & = & ((150+75)/2 \approx 112, -1) \\ (x_2, a_2) & = & ((112+75)/2 \approx 93, +1) \\ (x_3, a_3) & = & ((93+112)/2 \approx 102, -1) \\ (x_4, a_4) & = & ((102+93)/2 \approx 97, +1) \\ (x_5, a_5) & = & ((102+97)/2 \approx 99, +1) \\ (x_6, a_6) & = & ((102+99)/2 = 100, +1) \end{array}$$

and after trial 6, the producer's binary search converges to $w^* = 100$, and the producer will charge $p = 100$ forever from $t \geq 7$. In this case, the consumer's total profit is $25 + \beta^2 \cdot 7 + \beta^4 \cdot 3 + \beta^5 \cdot 1 + \beta^6 \cdot 0 + \sum_{i=7}^{\infty} \beta^i \cdot 0 = 25 + 7\beta^2 + 3\beta^4 + \beta^5$.

2.2 Noise Model

The base conjecture of this game is that randomisation obscures the signal so that the producer cannot observe the willingness to pay. This transforms the Baseline Model into a "guess with lies", and the producer side faces a noisy binary search problem.

Recall the deterministic baseline policy is $\bar{\pi}(a_t|x_t) = \mathbb{1}(a_t = +1)\mathbb{1}(x_t \leq 100) + \mathbb{1}(a_t = -1)\mathbb{1}(x_t > 100)$. A "lie" occurs when consumer deviates from it :

$$\begin{cases} x_t \leq 100 \text{ but } a_t = -1 & \text{rejecting a good offer} \\ x_t > 100 \text{ but } a_t = +1 & \text{accepting a bad offer} \end{cases}.$$

Under such mixed policy, compute the conditional lie probabilities :

$$\begin{cases} \mathbb{P}(a_t = -1 | x_t \leq 100) = \epsilon \cdot 0 + (1-\epsilon)(\frac{1}{2}) = \frac{1-\epsilon}{2} \\ \mathbb{P}(a_t = +1 | x_t > 100) = \epsilon \cdot 0 + (1-\epsilon)(\frac{1}{2}) = \frac{1-\epsilon}{2} \end{cases}.$$

Denote $\alpha = \mathbb{P}(x_t \leq 100)$. Then the lie rate is

$$
\begin{aligned}
p = \alpha\mathbb{P}(a_t = -1|x_t \le 100) + (1-\alpha)\mathbb{P}(a_t = +1|x_t > 100) \\
= \alpha\frac{1-\epsilon}{2} + (1-\alpha)\frac{1-\epsilon}{2} = \frac{1-\epsilon}{2}
\end{aligned}
$$

which implies the lie rate is independent of α, and therefore the noisy-search analysis becomes tractable.

Then based on noisy search theory [**Karp2007**], after t queries with lie rate p, posterior variance $\sigma_T^2 \ge C\exp(-T\cdot D(p))$ up to constant, where $D(p) = (1/2-p)^2$ is the discrimination rate. Then I have

$$\sigma_T^2 \ge C\exp(-T\cdot\epsilon^2/4) > 0$$

Therefore σ_T^2 is bounded below by a strictly positive constant depending on ϵ. This shows $\forall\epsilon > 0$, the producer's identification of w^* is slowed, ensuring positive expected consumer surplus over the infinite horizon. Explicitly, the consumer's value function is $V(x_t) = \mathbb{E}[(100-x_t)\mathbb{1}(a_t = +1) + \beta\cdot V_{t+1}]$ and $\mathbb{P}(a_t = +1) = \epsilon\mathbb{1}(x_t \le 100) + (1-\epsilon)/2$. This optimisation problem can be solved iteratively by the tabular Q-learning algorithm.

3 Empirical Results

For further development, I have a bellman equation with continuous state (price) and discrete action (accept/reject). I took three approaches; Monte Carlo Simulation, Tabular Q-Learning and Deep Q-Network.

3.1 Phase 1

In phase 1 (the first 0-6 trials), the producer plays the "take the middle" policy like they are in the "guess a number" game, while consumer randomises its behaviour with ϵ. Then we have a randomised version of state variable and consumer's behaviour as

$$
\begin{aligned}
(x_0, a_0) &= (75, \tfrac{1+\epsilon}{2}) \\
(x_1, a_1) &= (\tfrac{149+75\epsilon}{2}, \tfrac{1-\epsilon^2}{2}) \\
(x_2, a_2) &= (74.5 + 37.5\epsilon - 19\epsilon^2, \tfrac{2+\epsilon+\epsilon^3}{4}) \\
(x_3, a_3) &= (\tfrac{594+336\epsilon-150\epsilon^2+36\epsilon^3)}{8}, \tfrac{4+\epsilon-3\epsilon^2-\epsilon^3-\epsilon^4}{8}) \\
(x_4, a_4) &= (\tfrac{-9\epsilon^4+26\epsilon^3-180\epsilon^2+346\epsilon+593}{8}, \tfrac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16}) \\
(x_5, a_5) &= (\tfrac{5\epsilon^5-11\epsilon^4+68\epsilon^3-364\epsilon^2+703\epsilon+1183}{16}, \tfrac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16}) \\
(x_6, a_6) &= (\tfrac{7\epsilon^5-7\epsilon^4+76\epsilon^3-368\epsilon^2+709\epsilon+1183}{16}, \tfrac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16})
\end{aligned}
$$

As a result, we reached somewhat a mixed-strategy equilibrium and the value for consumer at this phase becomes

$$V(\epsilon,\beta) = \frac{-\beta^6\epsilon^5 + 45\beta^6\epsilon^4 + 38\beta^6\epsilon^3 - 462\beta^6\epsilon^2 - 37\beta^6\epsilon + 417\beta^6 + \beta^5\epsilon^5 + 49\beta^5\epsilon^4 + 46\beta^5\epsilon^3}{32} + \frac{-250\beta^5\epsilon^2 - 31\beta^5\epsilon + 417\beta^5 + 4\beta^4\epsilon^5 + 54\beta^4\epsilon^4 + 64\beta^4\epsilon^3 - 420\beta^4\epsilon^2 - 20\beta^4\epsilon + 414\beta^4 + 12\beta^3\epsilon^4}{32} + \frac{8\beta^3\epsilon^3 - 424\beta^3\epsilon^2 - 8\beta^3\epsilon + 412\beta^3 + 56\beta^2\epsilon^3 - 296\beta^2\epsilon^2}{32} + \frac{56\beta^2\epsilon + 408\beta^2 - 408\beta\epsilon^2 + 408\beta + 400\epsilon + 400}{32}$$

For the closed form solution as a function of β, I have the first order condition;

$$-\frac{5}{32}\beta^6 + \frac{45}{8}\beta^6\epsilon^3 + \frac{57}{16}\beta^6\epsilon^2 - \frac{231}{8}\beta^6\epsilon - \frac{37}{32}\beta^6 + \frac{5}{32}\beta^5\epsilon^4 + \frac{49}{8}\beta^5\epsilon^3 + \frac{69}{16}\beta^5\epsilon^2 - \frac{225}{8}\beta^5\epsilon - \frac{31}{32}\beta^5 + \frac{5}{8}\beta^4\epsilon^4 + \frac{27}{4}\beta^4\epsilon^3 + 6\beta^4\epsilon^2 - \frac{105}{4}\beta^4\epsilon - \frac{5}{8}\beta^4 + \frac{3}{2}\beta^3\epsilon^2 - \frac{53}{2}\beta^3\epsilon - \frac{1}{4}\beta^3 + \frac{21}{4}\beta^2\epsilon^2 - \frac{37}{2}\beta^2\epsilon + \frac{7}{4}\beta^2 - \frac{51}{2}\beta\epsilon + \frac{25}{2} = 0$$

but getting solution as a closed form requires unnecessary complexity. Instead, I plugged some different values of β in the above equation. As β differs from 0.5 to 0.99, ϵ^* varies from 0.57 to 0.07. Note the acceptance probability stabilises for $t \geq 4$ as the search localises near $x^* = 100$.

3.2 Phase 2

- **Monte Carlo**

  I tabularised the continuous states (producer's distribution) and apply Monte-Carlo iteration to compute the value function and optimal ϵ for consumer's profits. For computation simplicity, I could not use the phase 1 data, instead assume $\mathcal{N}(100, 10^2)$. Then producer updates his belief about willingness to pay by Bayesian update with the observation of consumer's behaviour, and setting the new distribution's mean as a new price.
  Here my pivotal argument is that if consumer is more forward-looking and therefore explore more, they earn more of surplus. It is derived by two results here. I first conjectured that myopic consumers (low β) prefer immediate optimal play over exploration, so ϵ should decrease with β. So I iterated with different time discount factor, and the result exactly shows the inverse correlation between them, the left of Figure 1. Also, when the consumer

Algorithm 2 Monte Carlo Simulation

```
t,V=0,0
  while t <= 200 do
    price=mean
    r =unif(0,1)
    if r < ϵ then
      a = +1 if price<=wtp else a = −1
    else
      a=unif({-1,+1})
    end if
    if a = +1 then
      V += β^t * (100 − price)
    end if
    update μ,σ via Bayesian rule given (price,a)
    t += 1
  end while
```

is more foresighted, in other words value more of future surplus, they get higher expected values, the right of Figure 1. These two empirical correlations confirm the theoretical hypothesis: patient consumers optimally trade immediate surplus for reputation, yielding higher long-run value.

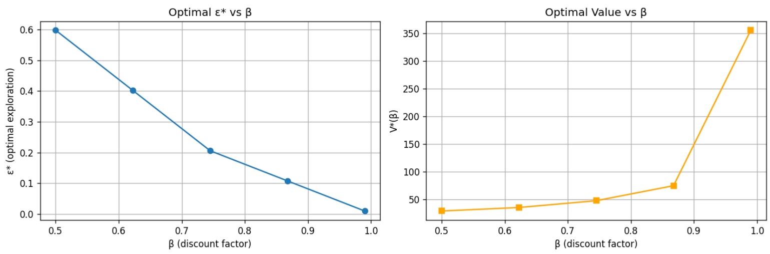

Figure 1: Correlations drawn from Monte Carlo

- **Tabular Q-Learning**

As set in section 1, producer's belief is $\mathcal{N}(\mu_t, \sigma_t^2)$ and producer charges $x_t = \mu_t$. As the name suggests, we discretise it onto the grid $A_0 = \{0, 1, \ldots, 150\}$ for tabular Q-learning so that the Q-table is indexed by observed states. Then we can compute posterior probability $\mathbb{P}(wtp = x' | data_t) = \mathcal{N}(x'; \mu_t, \sigma_t^2)$. The consumer applies Q-learning to learn optimal ϵ given observed prices. The Q-table implicitly captures expected future surplus through the Bellman backup. Firstly, define the Q-function for my problem

$$Q(x,\epsilon) = \mathbb{E}[u(x,\epsilon) + \sum_{j=1}^{\infty} \beta^j (100 - x_{t+j}) \mathbb{1}(a_{t+j} = +1) | x_t = x, \epsilon_t = \epsilon, \mu]$$

which has Bellman optimality equation

$$Q^*(x,\epsilon) = u(x,\epsilon) + \beta \sum_{x'} P_{xx'}(\epsilon) \max_{\epsilon'} Q^*(x',\epsilon'),\ \forall x \in X, \epsilon \in \{0.0, 0.1, \dots, 1.0\}$$

With above equations, I can execute Q-learning update

$$Q_{t+1}(x_t,\epsilon_t) = (1-\gamma)Q_t(x_t,\epsilon_t) + \gamma(u(x_t,\epsilon_t) + \beta \max_{\epsilon'} Q_t(x_{t+1},\epsilon'))$$

where the learning rate is denoted as γ.
For state transition, I sample a fixed number of price states from the producer's posterior distribution, and build the Markov transition matrix based on that. As the game is repeated, this Markov transition matrix evolves.

We conjecture that stationary play of the stage BNE constitutes a Markov perfect equilibrium of the discounted repeated game, with stationary joint distribution over $(\mu_t, \sigma_t^2, \epsilon_t)$. For the empirical validation, I simulate the game below. So even though the price state is not exactly Markov in the absence of producer's belief, we conjectured convergence under the BNE-induced stationary distribution.

With the exploration $\nu \neq 0$, and stationary state distribution (BNE conjecture), Q-learning algorithm empirically converges and therefore the above algorithm will give approximately true optimal policy. Also, here we only have the information about cumulative distribution function, not the point estimate, and therefore we use cdf to update the normal distribution (exact python code is provided at the github). The result gives no clear monotone correlation between time discount factor and optimal epsilon value. The optimal epsilon shows no monotone dependence on the time discount factor β. Here we impose the regularisation floor $\sigma > 0.1$ for numerical stability; the unregularised dynamics exhibit collapse, so the bounded-σ behaviour should be read as a consequence of this floor rather than an emergent property of randomisation.

- **Deep Q-Learning**

Here we also use the same setting with slight adjustments, we define the game. We define $i = 0$ (producer), $i = 1$ (consumer), with action spaces

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Algorithm 3 Tabular Q-Learning

Input: β time discount factor, initial normal distribution p_0, η convergence constant, T_{max} maximum iteration, learning rate γ, exploration rate ν.
Initialize Q_0 with zeros, $x_0 = \text{mean}(p_0)$, $t_0 = 0$.
while $t < T_{max}$ **do**
  Observe x_t and find $\epsilon_t^* = \arg\max Q_t(x,\epsilon)$
  With probability ν, play randomly from epsilon grid. With probability $1-\nu$, play ϵ_t^*.
  Sample $a_t \sim \pi_{\epsilon^*}(\cdot|x_t)$, where $\pi_{\epsilon^*}(a = +1|x_t) = \epsilon^* \mathbb{1}(x \le 100) + (1-\epsilon^*)/2$
  Observe $u_t(x_t,\epsilon_t^*) = (100 - x_t)\mathbb{1}(a_t = +1)$.
  update $\begin{cases} p_t \text{ with Bayesian update given } (x_t, a_t) \\ \qquad\qquad x_{t+1} = \text{mean}(p_{t+1}) \end{cases}$
  $Q_{t+1}(x_t,\epsilon_t^*) = Q_t(x_t,\epsilon_t^*) + \gamma(u_t(x_t,\epsilon_t^*) + \beta \max_{\epsilon'} Q_t(x_{t+1},\epsilon') - Q_t(x_t,\epsilon_t^*))$
  if $||Q_{t+1} - Q_t||_\infty < \eta$ **then**
    Break
  end if
  $t+ = 1$
end while
return Q, π^*

$A_0 = [0,150]$ (continuous price), $A_1 = [0,1]$ (continuous exploration rates), and stage payoffs:

$$u_0(x,\epsilon) = (x-50)\cdot \mathbb{P}(a = +1|x,\epsilon)$$
$$u_1(x,\epsilon) = (100-x)\cdot \mathbb{P}(a = +1|x,\epsilon)$$

$a \sim \pi_\epsilon(\cdot|x)$ where $\mathbb{P}(a = +1|x,\epsilon) = (\epsilon)\mathbb{1}(100 \ge x) + (1-\epsilon)/2$. We apply Reny's existence theorem[**Reny1999**] for discontinuous games.

Proposition 3.1 (Reny,1999)**.** *If $G = (X_i, u_i)_{i=1}^N$ is reciprocally u.s.c. and payoff secure, then it is better-reply secure.*

Theorem 3.1 (Reny,1999)**.** *If $G = (X_i, u_i)_{i=1}^N$ is compact, quasiconcave, and better-reply secure, then it possesses a pure strategy Nash equilibrium.*

For each i, let M_i denote the set of Borel probability measures on A_i. The mixed extension $\bar{G} = (M_i, \bar{u}_i)_{i=1}^N$ has payoffs $\bar{u}_i(\mu_1,\ldots,\mu_N) = \int_{A_1\times\cdots\times A_N} u_i(x_1,\ldots,x_N)\, d(\mu_1 \otimes \cdots \otimes \mu_N)$.

Corollary 3.1 (Reny,1999)**.** *Suppose $G = (X_i, u_i)_{i=1}^N$ is a compact Hausdorff game. Then G possesses a mixed strategy Nash equilibrium if its mixed extension $\bar{G}$ is better reply secure.*

Verification of Condition (Proposition 3.2): Here we define

$$\Phi_0(x) = \begin{cases} x & \text{if } x \in [0, 100-\nu] \cup [100+\nu, 150] \\ 100 & \text{if } x \in [100-\nu, 100+\nu] \end{cases}$$

where ν is arbitrarily small number > 0 for producer and

$$\Phi_1(\epsilon) = \epsilon$$

for consumer.

- producer

  for $i = 0$, $u_i(x_i^*, a'_{-i}) > u_i(x_i^*, a_{-i}) - \delta$
  $\Leftrightarrow u_0(x^*, \epsilon_1) - u_0(x^*, \epsilon_0) < \delta$
  $\Leftrightarrow (x^* - 50) \cdot (\mathbb{P}(a = +1|x = x^*, \epsilon_1) - \mathbb{P}(a = +1|x = x^*, \epsilon_0)) < \delta$
  $\Leftrightarrow (x^* - 50) \cdot [(\epsilon_1 - \epsilon_0)\mathbb{1}(x^* \leq 100) - \frac{1}{2}(\epsilon_1 - \epsilon_0)] < \delta$ hence $|u_0(x^*, \epsilon_1) - u_0(x^*, \epsilon_0)| \leq |K| \cdot |\epsilon_1 - \epsilon_0|$ where $K = (x^* - 50)[\mathbb{1}(x^* \leq 100) - 1/2]$ (here if $K = 0$, $u_0 = 0$, trivial) . Taking $V_{\epsilon_0} = \epsilon_1 : |\epsilon_1 - \epsilon_0| < \delta/|K|$ (or $V_{\epsilon_0} = A_1$ if $K = 0$) gives payoff security at x^*.

  But here we have a discontinuity at $x = 100$, which should be dealt separately. This means when $x^* = 100$, there is a jump in ϵ since $\mathbb{P}(a = +1|x = 100) = \frac{1+\epsilon}{2}$ while $\mathbb{P}(a = +1|x = 100 + \eta) = \frac{1-\epsilon}{2}$, $\forall \eta > 0$. So at $x = 100 + \eta$, where η is sufficiently small so that $x \in [100 - \nu, 100 + \nu]$. Then for a fixed ϵ_1 with varying ϵ_0 $\begin{cases} \text{LHS}: & u_0(100, \epsilon_0) = 50\frac{1+\epsilon_0}{2} \\ \text{RHS}: & u_0(100+\eta, \epsilon_1) - \delta = \frac{1-\epsilon_1}{2}(50 + \eta) - \delta \end{cases}$.
  When $\epsilon_0 = \epsilon_1$, $25(1 + \epsilon_1) \geq 25(1 - \epsilon_1) + \eta(1 - \epsilon_1)/2 - \delta \Leftrightarrow 50\epsilon_1 + \delta \geq \eta(1 - \epsilon_1)/2$, $\forall \epsilon_1 > 0$ with sufficiently small $\eta > 0$. By the continuity in ϵ, payoff secure follows.

- consumer

  for $i = 1, u_1(x, \epsilon) = (100 - x) \cdot \mathbb{P}(a = +1|x, \epsilon)$ is jointly continuous on $A_0 \times A_1$ (since (100-x) vanishes at $x = 100$, discontinuity in $\mathbb{P}$ is eliminated). Thus payoff security holds with $\Phi_1(\epsilon) = \epsilon$.

Reciprocal USC of sum : $u_0(\cdot) + u_1(\cdot)$ is continuous except at $x = 100$. Therefore, it is u.s.c. for those continuous intervals. At $x = 100$,
$\begin{cases} u_0(100, \epsilon) + u_1(100, \epsilon) = 50\frac{1+\epsilon}{2} + 0 = 25(1 + \epsilon) \\ \lim_{x \to 100^+} u_0 + u_1 = 50\frac{1-\epsilon}{2} + 0 = 25(1 - \epsilon) \\ \lim_{x \to 100^-} u_0 + u_1 = 50\frac{1+\epsilon}{2} + 0 = 25(1 + \epsilon) \end{cases}$. Therefore $\limsup_{x \to 100}(u_0 + u_1) = 25(1 + \epsilon) = (u_0 + u_1)(100, \epsilon)$, so the sum is upper semicontinuous at

$x = 100$. Elsewhere $u_0 + u_1$ is continuous, hence USC on $A_0 \times A_1$. This guarantees better-reply secure by the proposition 3.1. and therefore, we have the mixed strategy Nash Equilibrium for this game.

As set in the previous section, producer's belief is $\mathcal{N}(\mu_t, \sigma_t^2)$ and producer charges $x_t = \mu_t$. This is known to the consumer, so consumer observes x_t and computes μ_t, σ_t based on the previous data. This is similar setting to the original DQN paper[**Mnih2013**], where they observe raw pixels and process them into another state variable. For the DQN implementation, we discretise the consumer's action space A_1 to a finite grid $\{0, 0.01, 0.02, \ldots, 1\}$ of 101 values. Existence of mixed-strategy NE for the continuous game suggests the discretised game has approximate NE for sufficiently fine grids. Additionally, we adopt the standard ϵ-greedy exploration scheme; to avoid the notation conflict with the consumer's mixing parameter ϵ, we denote the exploration probability by ν throughout.

After executing simulations with 50000 iteration under the multiseed condition, we found out the result as in the Table 1. Running the original dynamics reveals a degeneracy: the posterior collapses to a point mass, so all post-collapse rewards vanish and the learned ϵ^* becomes insensitive to γ. Imposing a variance floor $\sigma > 0.1$ consistent with the Monte Carlo dynamics removes this degeneracy and recovers a significant monotone decrease in $\epsilon^*(\beta)$. Across all β, $\epsilon^* < 1$, confirming randomisation strictly improves consumer surplus over the deterministic policy.

Table 1: DQN-learned ϵ^* across discount factors (original dynamics, no variance floor, 5 seeds, $T = 50000$). The posterior collapses ($\sigma \to 10^{-3}$), so for $\gamma \in \{0.5, 0.7, 0.9\}$ all post-collapse rewards vanish and the learned policy is insensitive to γ; the rows coincide. Only the high-γ regime retains signal.

γ	ϵ^* (mean)	std	final σ
0.50	0.516	0.240	0.001
0.70	0.516	0.240	0.001
0.90	0.516	0.240	0.001
0.99	0.084	0.073	0.001

3.3 Ablation Studies

In terms of tabular Q-learning's discrepancy, we first thought that it may arise due to the discretisation of states. Thus, I add one more dimension, i.e. 2D state (μ, σ) and compare the result with the previous one. These two are statistically

Algorithm 4 Deep Q-Network

Input: β (discount), $T_{\max}$ (max iterations), γ (learning rate), ν (exploration rate), η (minibatch size), K (target update period), $|\mathcal{D}|$ (buffer capacity), initial prior p_0.

Initialize Q_Θ with random weights; target network Q_{Θ^-} with $\Theta^- \leftarrow \Theta$.

Initialize replay buffer $\mathcal{D}$ with capacity $|\mathcal{D}|$.

Initialize $s_0 = (\mu_0, \sigma_0)$ from p_0; $t \leftarrow 0$.

while $t < T_{\max}$ **do**

  Producer posts $x_t = \mu_t$.

  $\epsilon_t^* = \arg\max_\epsilon Q_\Theta(s_t, \epsilon)$.

  With probability ν, sample ϵ_t uniformly from the action grid; with probability $1-\nu$, $\epsilon_t = \epsilon_t^*$.

  Sample $a_t \sim \pi_{\epsilon_t}(\cdot \mid x_t)$ and observe $u_t = (100 - x_t) \cdot \mathbb{1}(a_t = +1)$.

  Producer's Bayesian update: $p_{t+1} \sim \mathcal{N}(\mu_{t+1}, \sigma_{t+1}^2)$ given (x_t, a_t).

  $s_{t+1} = (\mu_{t+1}, \sigma_{t+1})$; store $(s_t, \epsilon_t, u_t, s_{t+1})$ in $\mathcal{D}$.

  if $|\mathcal{D}| \geq \eta$ **then**

    Sample random minibatch $\{(s_j, \epsilon_j, u_j, s'_j)\}_{j=1}^{\eta}$ from $\mathcal{D}$.

    $y_j = u_j + \beta \cdot \max_{\epsilon'} Q_{\Theta^-}(s'_j, \epsilon')$.

    $\Theta \leftarrow \Theta - \gamma \cdot \nabla_\Theta \left[\frac{1}{\eta} \sum_j (y_j - Q_\Theta(s_j, \epsilon_j))^2 \right]$.

  end if

  if $t \bmod K = 0$ **then**

    $\Theta^- \leftarrow \Theta$.

  end if

  $t \leftarrow t + 1$.

end while

Return Q_Θ.

Table 2: DQN-learned ϵ^* across discount factors with a consistent variance floor $\sigma \geq 0.1$ (20 seeds, $T = 50000$). The Spearman rank correlation is computed over the raw (γ, ϵ^*) pairs (n=80): ϵ^* decreases significantly with γ, matching the Monte Carlo trend.

γ	ϵ^* (mean)	std	final σ
0.50	0.496	0.267	0.100
0.70	0.472	0.266	0.100
0.90	0.395	0.310	0.100
0.99	0.164	0.138	0.100
Spearman $\rho = -0.428$, $p = 0.0001$			

indistinguishable, and thus the conjecture that coarse 1D-discretisation collapses the capability is not supported.

Table 3: Tabular Q-learning ablation: ϵ^* under 1D state (μ only) versus 2D state (μ, σ), averaged over 20 seeds. The Spearman rank correlation is computed over the raw (β, ϵ^*) pairs (n=120). Enriching the state representation does not recover a monotone trend.

β	ϵ^* (1D, μ)	ϵ^* (2D, μ, σ)
0.50	0.352 ± 0.286	0.406 ± 0.278
0.60	0.332 ± 0.291	0.390 ± 0.286
0.70	0.362 ± 0.316	0.406 ± 0.310
0.80	0.258 ± 0.265	0.331 ± 0.283
0.90	0.380 ± 0.282	0.364 ± 0.297
0.99	0.496 ± 0.176	0.581 ± 0.187
Spearman ρ	+0.176	+0.128
p-value	0.054	0.164

4 Conclusion

All empirical results coincide in the sense that they all show the benefits of randomisation: optimal $\epsilon^* < 1$, indicating randomisation improves on deterministic play in the consumer-producer game.

The methods disagree on the precise function form of $\epsilon^*(\beta)$. Monte Carlo which optimises the exact phase-1 value, and DQN, which approximates the value over the continuous belief state, both recover a significant decreasing $\epsilon^*(\beta)$, while tabular Q-learning does not. We initially attributed this to coarse discretisation of the belief state, but the ablation studies rule this out. The likely cause is estimator variance on a flat objective - the consumer's value is shallow in ϵ so the tabular $\arg\max$ is dominated by seed noise - rather than of state representation.

This convergence of three independent methods (Monte Carlo, tabular Q-learning, and DQN) on the qualitative finding $\epsilon^* < 1$ with the quantitative agreement between MC and DQN, provides strong empirical support for randomisation as an optimal strategy in the obfuscation game.

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